Which Model Changes Can Counterfactual Explanations See? A Ground-Truth Benchmark

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Abstract. Counterfactual methods are used to explain how a classifier changed after an update, but without ground truth one cannot tell which method explains which kind of change. We inject known changes into classifiers on rendered images with exact factors, including a grid over the rotation and the offset of the decision boundary, and compare counterfactual differencing, gradient differences, importance differences, disagreement search and differential counterfactuals (Δ-CE), minimal inputs on which exactly one model changes its decision by a margin. A linear analysis predicts that gradient differences see a rotation and are blind to an offset; both halves were confirmed on CNNs in tests registered before the grid was run (Recovery 0.08 on offsets; +0.57 from the smallest to the largest rotation). Margin witnesses abstain on null updates, on rendered images and on a real tabular dataset, where counterfactual differencing attributes a change to about 99% of points; a differencing score thresholded to flag no null query also flags none below the witness threshold. After projection, witnesses and gradient differences differ when the old model's normal is taken at the query and nearly agree when it is taken at the witness (post hoc, confirmed on new data). A pre-registered hypothesis in favour of Δ-CE was rejected.

1 Introduction

Deployed classifiers are updated routinely, and an update that improves accuracy can still change individual decisions in unwanted ways [11]; explaining *how* they differ is a goal of explainable AI under distributional change [3, 12]. Artelt et al. [1, 2] compare a counterfactual [8] of the same input under each model, or approximate this by a difference of gradients; Muschalik et al. [4] compare feature importances; disagreement search [5, 6, 7] looks for inputs on which two models disagree. These families assume different things about *what* changed (a direction of the weights, a boundary, a region of disagreement), and without ground truth these assumptions go unchecked.

We ask which kind of change each family can see, using injected changes with exact ground truth. We also include *differential counterfactuals* (Δ-CE): the closest input on the data manifold at which only f_{new} changes its decision while f_{old} keeps it, both by a margin κ. Δ-CE anchors disagreement search [5, 6] at a query; unlike robust counterfactuals [9, 10], which seek a point valid for both models, it seeks one valid for exactly one, and the margin makes it return nothing for models that differ only by jitter.

Contributions. (i) A benchmark with exact ground truth for explaining model updates: five rendered factors, three injected changes, null updates, and a

4×4 grid over the rotation and threshold of the decision boundary (32 of the 49 model pairs), with code and protocol. (ii) A linear analysis separating rotation from offset. (iii) A test of these predictions on CNNs under a protocol fixed in advance. The linear analysis transfers to a local quantity (the gradient at the query) and not to a non-local one (the witness); tests that follow from a lemma or definition are separated from risky predictions in Table 3, and all failures are reported.

2 Differential counterfactuals and what each family sees

Let $D : \mathcal{Z} \to \mathcal{X}$ map a search space onto the data manifold (here an exact renderer from factor space $\mathcal{Z} = [0,1]^5$). For class y the margin is $m(z) = f_y(D(z)) - f_{1-y}(D(z))$; m_o, m_n belong to f_{old}, f_{new}, and $h = m_n - m_o$. A query z_0 is valid if $m_o(z_0) \geq \kappa$ and $m_n(z_0) \geq \kappa$.

Definition 1. With $\mathcal{F}_\kappa^N = \{z : m_n(z) \leq -\kappa,\ m_o(z) \geq \kappa\}$, a Δ-CE of z_0 is $z^\star \in \arg\min_{z \in \mathcal{F}_\kappa^N} \|z - z_0\|$ if $\|z^\star - z_0\| \leq r$, and NOT_FOUND otherwise; swapping the models gives type O. Each $z \in \mathcal{F}_\kappa^N$ is a *witness*: two forward passes verify that the models decide differently, both decisively, which a difference of two counterfactuals does not.

Lemma 1. Every witness has $h(z) \leq -\kappa - m_o(z) \leq -2\kappa$. Hence models whose margins differ by less than 2κ everywhere have no witness.

Linear case. For linear margins on $\mathbb{R}^p$ with independent w_o, w_n the nearest witness is unique: the ordinary counterfactual of f_{new} or the minimum-norm point with both constraints active. Write $\hat{w}_n = \cos\theta\, \hat{w}_o + \sin\theta\, v$ with $v \perp \hat{w}_o$, and let d_n, d_o be the distances from z_0 to $\{m_n = -\kappa\}$ and $\{m_o = \kappa\}$. The gradient difference of [1] is $-(\hat{w}_n - \hat{w}_o)$, counterfactual differencing is $u_n - u_o$, the difference of the two models' minimum-norm counterfactuals, the witness is $u^\star$.

Proposition 2. (a) *Offset* ($\theta = 0$): the gradient difference is 0; counterfactual differencing is a multiple of $\hat{w}_o$; a witness exists iff $d_n \leq d_o$ and equals $-d_n \hat{w}_o$. (b) *Rotation* ($0 < \theta < \pi/2$): the gradient difference has components in ratio $\tan(\theta/2) < 1$ along $(\hat{w}_o, v)$, i.e. mainly along the new dependence v; the witness has $|u^\star \cdot \hat{w}_o| = \min\{d_n \cos\theta, d_o\}$ and, for $\theta \leq \pi/4$, is dominated by v only if $d_o(\sin\theta + \cos\theta) < d_n$. After projecting out $\hat{w}_o$ all three coincide up to scale (in the spirit of Theorem 1 of [1], which expresses the cosine of two weight vectors through their counterfactuals).

On CNNs the last statement holds approximately only with $\hat{w}_o$ taken at the witness (Section 4.2). On linear models the solver's distances agree to 10^{-3} (the closed forms are KKT-certified). In words, gradient differences see a *rotation* and are blind to an *offset*; witnesses exist only on the side whose margin the offset lowers; for a constant logit offset such as the prior correction [13], they abstain exactly when the shift is below 2κ (Corollary 1 in the proofs). The analysis was formulated after the main experiment and before the grid was run.

Table 1: Injected changes; $k^\star$ is the ground-truth factor (none for the grid cell $(0, 0.5)$). [a]Two pairs have no $k^\star$ ($t = 0.5$) and two failed the offset check ($t = 0.53$), so four enter H5a.

Change	Adaptation data	$k^\star$	Pairs
Shortcut	colour correlated with label (0.99)	colour	3
Subgroup	labels flipped for size > 0.75	size	3
Threshold	$y = \mathbb{1}[\text{aspect} > 0.6]$	aspect	3
Null	resample / size $\sim U[0.5, 1]$, same rule	—	5 / 3
Grid, $\alpha = 0$	$y = \mathbb{1}[\text{aspect} + \alpha(\text{colour} - 0.5) > t]$, $t \in \{.5, .53, .56, .6\}$	aspect	8[a]
Grid, $\alpha > 0$	same, $\alpha \in \{.1, .25, .5\}$	colour	24

3 Benchmark and protocol

Data and models. Images are 32×32 ellipses rendered from five independent factors in $[0, 1]$: aspect (task factor), size, colour, background and horizontal position; labels are $y = \mathbb{1}[\text{aspect} > 0.5]$ with 5% noise; the renderer is exact, so the factor change of any explanation is read without a learned oracle. f_{old} is a small CNN (five seeds; two on the grid); f_{new} is f_{old} fine-tuned on 10 000 samples from one condition (Table 1). A pair enters only if its change is visible in decisions on validation data (null pairs: decisive disagreement ≤ 0.02). For the grid we measure the realised rotation $\hat{\alpha}$ and threshold $\hat{t}$ of each f_{new} by a logistic fit to its decisions, and analyse at these measured values; all 32 realised the intended rotation, while offsets below about 0.05 were not produced reliably.

Metrics and methods. Per pair: $N = 300$ valid test points. Recovery@r is the share of points with a valid explanation within the budget whose largest factor change is $k^\star$ (NOT_FOUND and ties count 0; $k^\star$ in Table 1). Recovery uses type-N witnesses only (pre-declared); a query whose class the update raises (Threshold, class 0) has none and counts 0. All counterfactual methods share one solver (Adam with a decaying penalty, then SLSQP), the same points and the same budget $r = 0.54$, set once on validation. We compare Δ-CE ($\kappa = 0.5$) with **CF-diff**, counterfactual differencing [1] ; **Grad-diff**, the gradient difference [1]; **Imp-diff**, a permutation-importance difference [4]; **DIG**, a line search along segments to training points, after [5]; **No-prox**, Δ-CE without the proximity term (the DeepXplore objective [6]); and **CF-new**, the ordinary counterfactual of f_{new} alone.

Protocol. Every test split served at most one confirmatory test, after all choices were fixed. Table 3 lists every test; the grid's manipulation check, exclusion rule and statistics were fixed after a pilot and before its test data; one analysis-code error was later corrected. Bootstraps are paired and stratified by pair; they resample points within three pairs (two seeds of f_{old} on the grid), so intervals do not cover model-to-model variation, which we did not measure beyond re-solves of the same pair. The rest is descriptive. A digit tier and a label-shift update failed qualification and were stopped.

Table 2: Recovery@r (found-rate) with type-N Δ-CE, locked test split, mean over three pairs (95% CIs within ±0.035); last row: found-rate on the 8 null pairs.

	Δ-CE	CF-diff	Grad-diff	Imp-diff	DIG	No-prox	CF-new
Shortcut	0.38 (0.84)	0.54 (0.93)	**0.91** (1.0)	0.00	0.02 (0.08)	0.35 (0.63)	0.07 (0.93)
Subgroup	0.53 (0.81)	0.32 (0.99)	**0.62** (1.0)	0.00	0.06 (0.10)	0.29 (0.58)	0.30 (1.0)
Threshold	0.47 (0.47)	0.76 (0.90)	0.07 (1.0)	**1.00**	0.07 (0.08)	0.31 (0.40)	0.90 (0.90)
Null	**0.000**	0.994	1.0	1.0	0.000	0.000	0.996

Table 3: Tests (95% CIs; †90%). K: D = follows from a lemma, proposition or definition, R = risky prediction. Formed before any result: H2a, P-tie; after the first: C1–C3; before the grid: H5a–H5d; later (new data): H6–H8. Holm: {C1, C3}, {H5a–H5d}, {H7c, H8}. Proj.: $\hat{w}_o$ projected out.

Claim	Data	Estimate	p_{Holm}	Status	K	ID
Δ-CE > Grad-diff	Subgroup	-0.09 [-0.13, -0.05]		rejected	R	H2a
Tie ($\lvert\Delta\rvert \leq 0.1$)	Shortc., Thresh.	$\lvert\Delta\rvert = 0.17$–$0.53$		not conf.	R	P-tie
Δ-CE > Grad-diff	Threshold, new	$+0.31$ [0.27, 0.35]	$2{\cdot}10^{-4}$	confirmed	D	C1
No witness on nulls	Null; Adult	0/2 400; 0/1 500		confirmed	D	C2, H6
Proj. Δ-CE ≈ Grad-diff	Shortcut, new	-0.30 [-0.33, -0.28]†		not conf.	R	C3
Grad-diff < 0.2, offsets	Grid, $\alpha = 0$	0.081 [0.066, 0.096]	$4{\cdot}10^{-4}$	confirmed	R	H5a
Grad-diff rises with α	Grid	$+0.57$ [0.53, 0.61]	$4{\cdot}10^{-4}$	confirmed	R	H5b
Grad-diff offset-free	Grid	-0.098 [-0.143, -0.053]†		not conf.	R	H5c
Proj. Δ-CE ≈ Grad-diff	Grid	-0.35 [-0.38, -0.33]†		not conf.	R	H5d
CF-diff flags $\lvert b\rvert < 2\kappa$	Logit offsets	0.961 [0.955, 0.967]	$2{\cdot}10^{-4}$	confirmed	D	H7c
Witness normal narrows gap $\geq \frac{1}{2}$	Grid	$-0.19 \to +0.03$ [0.01, 0.05]†	$6{\cdot}10^{-4}$	confirmed	R	H8

4 Results

Findings are grouped by the kind of change; Table 3 lists every test.

4.1 Offsets

Gradient differences are blind to an offset, as Proposition 2(a) predicts: on the pure offsets of the grid their Recovery is 0.081, and on Threshold Δ-CE beats them (0.41 vs. 0.10 on the new split; Table 2 is the locked split). On an offset the changed factor is the task factor, so even CF-new recovers it (0.90–0.98). Type-N witnesses exist only on the side the offset lowers (on Threshold: every class-1 query, none of class 0), which is why Δ-CE scores 0.47 in Table 2; type O covers the other side. On new data (three Threshold pairs, 900 queries, descriptive) type N finds a witness for every class-1 query and none of class 0 (Recovery 0.45), type O the converse, and the nearer of the two recovers aspect on all 900 queries (1.00; CF-diff 0.77, CF-new 0.89, Grad-diff 0.07). For an exact logit offset b, witnesses appear only once $\lvert b\rvert$ reaches 2κ and then, empirically, on every query of the lowered class. CF-diff, with no threshold, flags 0.961 of queries already at $\lvert b\rvert \in \{0.5, 0.9\}$, where 3–8% of decisions do flip: that change is real and the witness misses it by design (Corollary 1), a difference in sensitivity, not an error

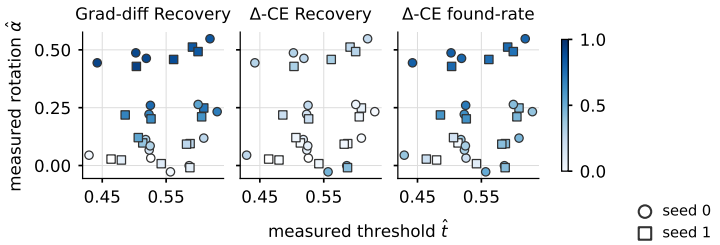


Fig. 1: Grid results at the measured rotation $\hat{\alpha}$ and threshold $\hat{t}$ of each f_{new} (test split), on one colour scale from 0 to 1. Hollow: no $k^\star$; marker shape: seed of f_{old}.

of CF-diff.

4.2 Rotations

Gradient differences see a rotation, increasingly with its angle: Grad-diff Recovery rises from 0.36 at $\alpha = 0.1$ to 0.93 at $\alpha = 0.5$ (Fig. 1), while CF-new recovers colour on ≤ 0.03 of points. Proposition 2(b) predicts that after projecting out $\hat{w}_o$ witnesses and gradient differences coincide up to scale; with $\hat{w}_o$ taken at the query they do not (Shortcut 0.64 vs. 0.95, all valid points; grid -0.35). A post-hoc explanation, tested on new data (four grid pairs, points with a witness): with $\hat{w}_o$ taken at the witness the projected gap Δ-CE − Grad-diff moves from -0.190 to $+0.029$. This locates the failure, not its cause. Descriptive: on Shortcut, on points with a witness, it moves from -0.166 to -0.024; the normal turns by a median 32° from z_0 to $z^\star$ and the registered prediction that this grows with $\|z^\star - z_0\|$ failed; the grid gap of -0.35 splits into -0.18 abstention and -0.17 geometry.

4.3 Null updates and abstention

Abstention comes from verifying the margin, which all three disagreement searches do: none returns a point on a null update (Table 2). What Δ-CE adds is the witness closest to the query: No-prox finds witnesses on fewer points (0.40–0.63 vs. 0.47–0.84) and, where both find one, the Δ-CE witness is nearer on every point (median 0.20–0.23 vs. 0.31–0.33; descriptive), with higher Recovery on Subgroup and Threshold. On the 8 null pairs no witness exists, while CF-diff attributes a change to 99.1–99.8% of points. Much of this contrast is the threshold. Scoring a query by $\|u_n - u_o\|$ and cutting on the 2 400 null queries so that none is flagged, as for Δ-CE, CF-diff flags none of the logit-offset queries for $|b| \leq 1.5$ and 0.001 at $|b| = 2$, where Δ-CE has a witness for every lowered-class query from $|b| = 1.1$; at 5% false positives it flags 0.17, 0.40 and 0.76 of the lowered-class queries at $|b| = 1.1, 1.5, 2$ (descriptive, saved outputs). With $\kappa = 0$ witnesses reappear (0.80, 0.89), as Lemma 1 predicts. The same holds for five null updates of an MLP on the numerical Adult features (no witness on 1 500 test points, CF-diff 97.7–100%). NOT_FOUND is not a solver failure: 10^5 sampled points per query

find no witness on 400 null and 150 impossible-class queries (95% bounds 0.9%, 2.4%) and miss one on 3 of 395 small-offset queries (descriptive).

What failed. On Subgroup Δ-CE is below Grad-diff (-0.091, negative in all seeds), rejecting the pre-registered hypothesis; it beats CF-diff ($+0.21$) and DIG ($+0.47$). The predicted tie on Shortcut and Threshold fails in all four comparisons, and adding an offset to a rotation lowers Grad-diff Recovery by 0.098 (90% CI $[-0.143, -0.053]$, not within ± 0.10).

5 Discussion and limitations

Witnesses are meant to show *where* decisions changed near a query, a notion Recovery does not score. **Limitations.** (1) The image data are synthetic; the tabular test covers abstention only. (2) Of label shift, only the exact prior correction [13] was tested. (3) κ is an exact threshold only for a constant offset; otherwise abstention tracks decisive change (at shifts of 0.03–0.04 it depended on the fine-tuning recipe). (4) Witnesses see only decision changes on the manifold within the budget, with no certificate for deep networks. (5) Recovery scores one notion of change, the factor along which the weights moved, the view of gradient differences; on an offset the task factor is the answer for any method that points at it. (6) Several tests were formed after earlier results, one after a validation run showed the effect (Table 3). Every protocol change was logged, with dated commits, before the test data it concerns were generated (the analysis-code correction excepted); log, proofs and code: [repository link].

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